Sai Venkata Karthik Bachu

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Profile

Software Engineer with hands-on full-stack and applied-AI experience from a live healthcare platform – building backend services, a real-time chat feature, and an LLM-powered booking assistant, alongside an end-to-end QA automation suite. Proficient across React, React Native, Node.js/Express, and Python/Java, with a strong foundation in Data Structures, Algorithms, and Machine Learning.

Education

Vellore Institute of Technology – AP 2022 – 2026
B.Tech in Computer Science and Technology
CGPA: 8.55 / 10.0

Resonance Junior College, India 2020 – 2022
Intermediate (Class XII)
Percentage: 86%

St. Joseph's School, India 2010 – 2020
ICSE – Class X
Percentage: 92%

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL
Frontend: React, React Native, Next.js, Three.js
Backend: Node.js, Express.js, Flask, REST APIs, Socket.io, JWT Auth
AI/ML: LLM Integration (Google Gemini, LangGraph.js), Agentic Workflows, PyTorch, Stable Diffusion (SDXL), Computer Vision (MediaPipe)
Databases & Cloud: MongoDB, MySQL, PostgreSQL, AWS (S3, EC2), Firebase
Testing & Tools: Playwright, Allure, Selenium, Git, SonarCloud, Excel

Experience

Software Engineering Intern Feb 2026 – Present
Vydhyo Healthcare

- Built an agentic AI doctor-booking assistant (Express.js, LangGraph.js, Google Gemini 2.5 Flash) serving a 150-doctor/20-clinic dataset across 8 specialties and 10 cities, with Zod-validated tool calls, self-healing retries, and dual token-bucket rate limiting; shipped the React 19 + Vite chat widget.
- Developed real-time blood-donation request and chat features (Socket.io) in the patient app, including live request tracking, geocoding-based donor matching, and push notifications.
- Built and maintained a 560+ test Java/Playwright automation suite (TestNG, Allure reporting) across 40+ test classes spanning appointments, billing, e-prescription, and security flows.
- Ran REST API contract audits across 20 backend service groups and security/performance/load testing (k6, Locust), alongside QA tracking of 550+ test cases across 100+ feature modules.

Projects

Karthik's Pit Wall – F1 Dashboard React, TypeScript, Vite

- Built a real-time Formula 1 dashboard with live standings, constructor progress tracking, and a championship predictor, sourcing live data from the Jolpica/Ergast API.

- Implemented the UI with Radix UI primitives, TanStack Query for data fetching/caching, and Framer Motion animations; set up a SonarCloud CI pipeline and deployed on Vercel.

Ultron Orb UI Next.js, Three.js, MediaPipe

- Extended an open-source Three.js/MediaPipe hand-tracking interface with new features: an interactive 3D Earth globe, a HUD dashboard, a spatial sound engine, and live weather integration.
- Rewrote the webcam-based gesture-detection pipeline for more reliable pinch-to-spin and two-hand zoom controls.

Capstone-Flask – SDXL Image Studio Python, Flask, PyTorch

- Co-built a browser-based Stable Diffusion XL studio (Flask + PyTorch) supporting text-to-image generation, image-to-image transformation, and 4x upscaling via the SD x4 Upscaler model.
- Optimized GPU-accelerated inference with CUDA/xFormers (plus Apple Silicon MPS support), serving multi-gigabyte diffusion models through a lightweight, responsive web interface.

Encrypted Chat Application Java, AES Encryption

- Built a secure multi-client chat platform from scratch using raw TCP sockets and multi-threading to handle concurrent real-time connections.
- Integrated AES encryption for end-to-end message security, keeping conversations private even over an untrusted network.

Achievements

- Secured Top 15 position in a 24-hour hackathon among 60 teams.
- Published a research paper on Cross-Cultural Communication and Interaction.

Certifications

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional**
- Coursera – Introduction to Data Analytics
- Coursera – Data Visualization with Excel & Cognos
- Edureka – Selenium Automation Testing